33NATIONAL UNIVERSITY OF SCIENCE AND TECHNOLOGY

FACULTY OF COMMERCE

DEPARTMENT OF ACCOUNTING

MANAGEMENT ACCOUNTING II

CAC 4105

FIRST SEMESTER EXAMINATION PAPER

FEBRUARY 2022

This examination paper consists of 12 pages

Time Allowed: 3 hours

Total Marks: 100

Special Requirements:

Examiner's Name: Dr D. Madzivanyati

INSTRUCTIONS

1. Attempt all questions, the question paper consists of 4 questions.
2. Use the examination answer booklet provided
3. Use of calculators is permissible
4. Pay particular attention to the exact wording of the **required section**, and the number of marks allocated. Answers should be carefully planned before being executed and clearly numbered. The length and extent of answers should relate to the allocated number of marks.

MARK ALLOCATION

QUESTION	MARKS
1	25
2	30
3	20
4	25
TOTAL	100

Copyright: National University of Science and Technology, 2022

Question 1 – CVP ANALYSIS

Hotdog Eats (Pvt) Ltd (Hotdog) is a franchisor and wholesaler which recruits and selects individuals or businesses and offers them a franchise agreement. Currently there are 75 franchise holder's country-wide, most of which are independently owned and operated. These are operated as small to medium sized stalls which sells Hotdogs and complementary food stuff.

The franchise model is simple and Hotdog has sold 10 franchises in the past 12 months. They each sign a 5-year franchise agreement with the option to renew for another 5 years. After 'in-shop' training for two weeks, they receive a conversion kit and assistance to set up their own stall. In turn for a royalty fee they receive continual marketing and administrative support. The contract also stipulates that all product and consumables are to be purchased from the wholesale division of Hotdog as Hotdog aims to ensure quality and a consistent corporate identity.

Customers pay for the product in cash due to the nature of the products. Each franchise is required to report all sales on a monthly basis to Hotdog as part of the franchise agreement. The cash is banked an average once a week by each individual franchise. The Managing Director (MD) is however concerned as the reported average sales for the franchises are going down. Currently the regional manager from Hotdog visits each franchise twice a year.

Hotdog Franchise

Each franchise operates out of a movable stall situated in or near a shopping mall. The main product line-up consists of two types of Hotdogs: Easy-Dog with only a Vienna and complementary sauces and Ultra-Dog which includes cheese sausage and additional garnish topped with a secret sauce. It sells for $25 and $35 respectively. The ingredients are obtained from the Supply Division of Hotdog.

The MD tasked the junior accountant to calculate the break-even sales for a typical franchise, and she presented the following data, which excludes any tax effect:

Cost data for franchise-holders per unit of product	**Easy-dog**	**Ultra-dog**
Selling price	$25	$35
Packaging	$0,25	$0,5
Labour & variable Overhead (% of selling price)	5%	6%
Mark-up (on cost price paid to **Supply division**)	50%	60%
Royalty (payable on Revenue)	5%	5%

Other relevant information linked to a typical franchise-holder:

- Fixed overhead amounts to $250 000 per annum.
- Administrative cost allocated from Hotdog is $10 000 per month.

Copyright: National University of Science and Technology, 2022

- To finance a stall, at cost of $150 000, a franchise has to take out a loan from a preferred financial institution. On average the loan is repayable over three years at 10% interest p.a. compounded monthly.
- According to historic data it is expected that for each 2 Easy-dogs, 1 Ultra-dog is sold.

10 % interest

BEP = IFC contribution

Interest = 10% × 150 000

QUESTION 1			
REQUIRED		Marks	
		Sub-total	Total
(a)	Identify risks linked to Hotdog's franchise model and for each risk suggest how to mitigate the risk effectively.	(10)	(10)
(b)	Calculate the number of each type of hotdog a franchise must sell per month in order to break-even.	(15)	(15)
Total for Question 1			25

Copyright: National University of Science and Technology, 2022

QUESTION 2 – DECISION MAKING

PART A

The Bulawayo City Council is hosting a cultural day for schools in the City at ZITF Grounds and has asked Hotdog to prepare a quote to provide each child with a food package consisting of One Easy-dog Hotdog, One Soft drink, One Health bar. For the tender, the MD will assume a price for the Easy-dog of $12,50 from the Supply Division. The MD is eager to get the contract as there are many of these events country wide and he had a few ideas to maximise his profit.

- 350 children are expected to attend. As Hotdog wants to promote its brand, it said it will provide 10% additional packages at cost to cover for any additional children attending as well as free delivery.
- Soft drinks will have to be acquired. The current market price is $5,50 each. There are currently 150 cans in inventory which was acquired at $5,20 to send to the franchise-holders.
- Hotdog plans to provide an all-in-one parcel to the children by packing everything into a polystyrene box. Since this differs from their normal inventory, it will have to be acquired from a new supplier. There is however a minimum order quantity of 400 at 50c per box
- The health bars are also not sold at the stores and will have to be acquired at $4,50 each. The minimum order quantity is 2 000 bars. The MD suggested that they sell the rest of the bars at their Bulawayo stalls as an experiment.
- For packing and quality control of the parcel, workers will have to work overtime. The MD will have to work 20 additional hours to prepare the tender. His currently hourly rate is $450. Three wage earners who normally earn $35/hour will work 4 hours' overtime each for two days. The overtime rate is calculated at 1,5 times the normal wage rate. The Supply division are expected to work 10 hours' overtime in preparation of the Easy-dogs. These employees earn the same wage rates as the other wage labourers. Hotdog subscribes to the Basic Conditions of Labour/Employment act.
- The MD has found that this will be the ideal time to buy a new delivery vehicle he was planning on for a while now to help with operations at a cost of $285 000. ZIMRA has allowed an annual 20% wear and tear deduction.
- Fixed overhead is allocated at a rate of $20 per labour hour.
- Variable overhead is expected at $2,50 per parcel.
  The MD expects a 20% mark-up on relevant cost.

Copyright: National University of Science and Technology, 2022

PART B – DIRECT COSTING

Barcelo's was started by two Italian twin brothers, Kelvin and Keith Barcelo, who migrated to Zimbabwe to start an Italian take-away. After struggling for years in the very competitive market, Kelvin decided to use his grandfather's lasagna recipe and convert it into a microwavable meal, which can be sold to larger outlet stores such as Spar, Choppies and Pick N Pay. Barcelo's started off with the Julio, a meat lasagna but the Carrie, a vegetarian lasagna (named after his daughter) was recently introduced by Kelvin as market research showed that career women with children, are looking for a healthy meal that is easy to prepare. Barcelo's currently operates a small delicatessen in Bradfield and supplies a few independent grocers on a cash on delivery basis.

Even though it is an age-old recipe, production takes place at a small fairly automated plant. The production process starts with the mixing of ingredients and the kneading of the dough. The two lasagnas are then split into two processes, the meat and the vegetable process. The existing costing system is not very sophisticated as Barcelo's grew a bit too fast for the systems. The costs include direct materials (dough, butter, mince, vegetables). As Kelvin (with the help of his engineering friend) was able to convert the process into a fairly automated process, fixed production overheads are quite high. Kelvin is however confident, that should capacity increase, Barcelo's plant will be able to handle it.

Extracts from the budget for the year ended 30 September 2021 and the actual results per the management accounts for the year are set out below

BARCELO'S (Pvt) LTD

BUDGETED AND ACTUAL RESULTS FOR THE YEAR ENDED 30 SEPT 2021

	Notes	Budget	Actual
		$	$
Revenue	1	4 015 000	4 032 000
Less: Cost of Sales		(2 225 837)	(2 272 871)
Opening Inventory	2	297 000	297 000
Production costs	3	2 037 300	2 083 015
		2 334 300	2 380 015
Closing Inventory	4	(108 463)	(107 144)
Gross Profit		1 789 163	1 759 129

Copyright: National University of Science and Technology, 2022

Selling and administrative costs	5	(1 750 000)	(1 755 000)
Net profit before tax		39 163	4 129

Notes:

1. Budgeted selling prices are $50 for Julio and $45 for Carrie, but Kelvin believed that he could keep up with the budgeted sales volume (65 000 for Julio and 17 000 for Carrie) if he increased the selling price of the Carrie by $1 and was proved correct by the actual outcome. The Julio is currently producing at 80% capacity and the Carrie at 40%.
2. Opening inventory at 1 October 2020 consisted of 8000 units of Julio and 3 000 units of Carrie at an actual cost of $216 000 and $81 000 respectively. A fixed manufacturing expense per lasagne was $16 for the 2020 financial year. No material inventories of raw materials were held on this date.
3. Kelvin budgeted for 60 000 Julios and 15 000 Carries to be manufactured during the year. Actual production showed 59 000 Julios and 16 000 Carries were manufactured.
4. Barcelo's uses the Weighted Average method of inventory valuation and the closing inventory values on 30 September 2021 are considered to be correct
5. Selling and administrative cost consist of 60% fixed cost and 40% variable cost and a 6:1 ratio between Julio and Carrie. Variable cost was $35 000 more than budgeted.

Additional information:

- Flour, butter and dough are bought premixed from a local supplier. The budgeted cost per kg mix is $14. One Julio requires 220g and one Carrie 200g of the mix. Total actual cost of the mix for the year was $230 040 ($191 160 for Julio) after Kelvin was able to negotiate a price of $13,50 per kg.
- The actual mince cost for the Julio lasagna was $283 200. Kelvin was very concerned, because although the mince price remained at the budgeted $20 per kilogram this was $43 200 more than that was budgeted.
- Currently the vegetable portions that fill the Carrie are the same quantity that fills a Cornish pie. Kelvin budgeted $2 per Carrie lasagna. The actual cost was $30 400.
- As the manufacturing process is fairly automated, only six (6) unskilled workers are required. As they are willing to work for the minimum stipulated wage, Kelvin decided to keep them on. The workers are split in two groups, four (4) workers allocated for the Julio and the rest for the Carrie. Kelvin budgeted for 9 000 hours at the minimum rate of $15 per hour. Kelvin reviewed the timesheets and discovered that it takes on average 6 minutes for a Julio and 12 minutes for a Carrie to be processed. This is based on a 125-hour month per worker. Barcelo's were part of the striking members in a nationwide strike, the outcome of which forced Kelvin to increase their hourly rate by 50c. Keith is in control of the workers, and she allowed an extra 2 250 hours that Kelvin did not budget. The Julio workers clocked 8 856 hours in total and the Carrie workers 1 200 hours each.
- Budgeted variable manufacturing overheads were calculated at $2,50 per lasagna. Actual variable manufacturing overheads amounted to $180 000.

Copyright: National University of Science and Technology, 2022

- The lasagne is wrapped in a disposable labelled plastic holder which was budgeted for at 24c per holder. Due to increasing pressure from the public to go "green", the supplier offered a discount of 4 cents should Barcelo's continue with the order and pay cash, which Kelvin accepted.
- Fixed manufacturing overheads of $1 200 000 were budgeted. Currently these overheads are allocated based on produced quantities. Actual fixed manufacturing overheads amounted to $1 170 000.

QUESTION 2			
REQUIRED		**Marks**	
		Sub-total	**Total**
(a)	Calculate the expected selling price per food package for the special order linked to the Cultural Day on a relevant costing basis. Clearly show the relevant cost value for each of the items. Explain each relevant value you have evaluated and why the values you have excluded are not relevant. Discuss whether it is likely that the municipality will accept the deal.	(8) (2)	(10)
(b)	Draft a Budgeted Statement of Comprehensive Income for the year ended 30 September 2021 for both lasagnas applying direct costing principles. A total column is not required.	(15)	(15)
(c)	Indicate why you would prefer a variable costing system to an absorption costing system from a management accounting perspective.	(5)	(5)
Total for Question 2			**30**

Copyright: National University of Science and Technology, 2022

QUESTION 3 – STANDARD COSTING

Bela-Bela Cleaning Products (Pvt) Ltd ('BBCP') is a company situated in Dulibadzimu, a town in Beitbridge, and manufactures and distributes various cleaning products.

BBCP Soap Manufacturing Division

This division manufactures and supplies environmentally-friendly bars of soap to leisure, nature and wildlife lodges for use in their chalets and various facilities. The bars weigh 100g each and are sold for $5,50 each. They are sold in bags containing 10 bars of soap.

selling price $ 6,50

The Soap Manufacturing Division of BBCP manufactures the soap using an entirely automated production process. Supervision and maintenance costs fall under overhead.

The following information regarding the cost of a bag of 10 soap bars is available:

Input at standard cost:

0,30kg	of Ingredient C	at $38,00 per kg
0,65kg	of ingredient T	at $13,00 per kg
0,2kg	of Ingredient A	at $21,00 per kg
1,15kg		

Other information:

Variable overhead: 0,25 machine hours
Fixed overhead: 0,25 machine hours
Packaging: one (1) plastic bag at $0,30 per bag

Overheads are absorbed on the basis of machine hours. BBCP uses an absorption costing system.

The following budget information is available regarding total overhead costs (i.e. total variable overhead cost plus total fixed overhead cost) at different production levels (based on variable costing principles):

Copyright: National University of Science and Technology, 2022

QUESTION 3

$VC = \frac{31\,075 - 30\,220}{2\,500 - 2\,320}$

$VC/unit = \$\ 4{,}75$

FC

Total cost = FC + V.C

30 220 = FC

Month	Production Level Bags of Soap	Total overhead cost (fixed + variable) $
1	2 320	30 220
2	2 500	31 075
3	2 200	29 650

4,75

11 875

19 200

The January 2022 budget includes 600 machine hours and the production of 2 400 bags of soap, which was the most common production level of the past two years.

The actual results for January 2022 show the following:

- 2 490 bags of soap were produced.
- 750 kg of ingredient C was purchased at $39,00 per kg. 730 kg was issued to production.
- 1 550 kg of ingredient T was purchased for $20 460. 1 600 kg was issued to production.
- 560 kg of ingredient A was purchased at $18,50 per kg. 550 kg was issued to production.
- Variable overhead cost of $12 864 was incurred for 640 machine hours.
- The actual fixed overhead incurred was $20 000.
- Plastic bags were purchased at $0,30 per bag. The total cost amounted to $747

Inventory of raw materials, packaging and finished goods are valued at standard cost. The standard costs have not been changed for the past six months of 2021.

Copyright: National University of Science and Technology, 2022

REQUIRED		Marks	
		Sub-total	Total
(a)	Calculate the standard cost of one bag of soap bars for Bela-Bela Cleaning Products (Pty) Ltd (BBCP) using an absorption costing basis.	(5)	(5)
(b)	Calculate the following standard costing variances for BBCP's soap division for the month of January 2022: • Material price variances for ingredients C, T and A. (3) • Material mix and yield variances for ingredients C, T and A. (8) • Variable overhead expenditure variance and efficiency variance. (2) • Fixed overhead expenditure variance and volume variance. (2)		(15)
Total for Question 3			**20**

Cost card for One bag of soap
unit – 2440

LEY = SY
LRY = (SA – AA) AH

Copyright: National University of Science and Technology, 2022

QUESTION 4

QUESTION 4 – APPLICATION OF MARGINAL COSTING PRINCIPLES.

The Floor Cleaner Division manufactures three types of floor cleaner, namely Product F, Product L and Product O, which are sold in 750ml bottles. Details of the products are as follows:

Product:	F $/bottle	L $/bottle	O $/bottle
Selling price	30,00	22,00	26,50
Liquid A ($16 per litre)	4,80	3,20	5,60
Liquid B ($14 per litre)	4,20	3,50	2,80
Other ingredients and packaging	5,00	3,50	4,00
Variable overhead ($140 per machine hour)	4,20	2,80	4,20
Fixed overhead ($160 per machine hour)	4,80	1,60	1,60
Manufacturing profit	7,00	7,40	8,30

The machines used in this process can only be utilised for this particular manufacturing process. The demand for the three products for February 2022 has been estimated as follows (excluding fixed contracts):

	F	L	O
Bottles	1 000	2 100	1 500

In addition to the demand above, fixed contracts with lodge clients have been signed for the following quantities for February 2022:

No inventory

	F	L	O
Bottles	400	500	460

The Floor Cleaner Division does not hold any inventory of raw materials or finished goods.

The management of BBCP has advised that the availability of Liquid A and Liquid B will be restricted during February 2022 to 1 800 litres of Liquid A and 1 200 litres of Liquid B, as advised by the only local supplier of these liquids. BBCP has no intention of venturing into the import market for these liquids. No restrictions are expected on any other resources.

Available A 1 800
B 1 200
1 050 1 950

Copyright: National University of Science and Technology, 2022

REQUIRED		Marks	
		Sub-total	Total
(a)	Calculate the optimum product mix to maximise BBCP's floor cleaner division's profit for February 2022. Also calculate the number of bottles of product that cannot be supplied due to the shortage of Liquid A and/or Liquid B.	(18)	(18)
(b)	Briefly list BBCP's major risks in terms of liquids A and B and then explain why, in your opinion, it is reasonable for BBCP to import these liquids. [Risks: 3 marks, import acquisition 4 marks]	(7)	(7)
Total for Question 4			**25**

End of Examination Paper

Copyright: National University of Science and Technology, 2022